

issues which probably did exist in a miniature form, tend to bloat up in the cloud. Classic examples of this are:

- **Known architecture/design issues**, which are less of a threat to the overall operations as of date are often left unaddressed and parked aside for future upgrades. You end up taking known architectural and design anomalies to the cloud, and this is only going to multiply in the highly elastic computing environment.
- **Session management tactics adopted for a Web workload**: If you have a Web application that managed sessions in-proc, when you migrate the same workload and configure auto-scale capability, you will realise anomalies unless you set sticky bit for the load balancer.
- **A two-tier application where lack of intermediary services** leads to direct calls being made by the app to the database, which for security and regulatory purposes, continues to remain on premise.

The above examples may appear trivial, but do demand redesigning. In certain scenarios, we have observed that issues have only bubbled up post migration to the cloud when one has taken a stance to move over without assessing the true readiness of a particular solution.

Hence, it is highly recommended to assess the building blocks of the solution for its fitment and adaptability to the compute environment in the cloud. In addition, fix the existing issues in the realm of architecture or design before marching ahead. Otherwise, you are only taking the issues along to an environment where they can pose greater threats to the overall functioning of the solution.

2. Don't consider the cloud to be some sort of saviour that addresses performance

While the cloud gives unlimited access to compute, don't embrace it just to solve performance issues. This is similar to customers hosting their solutions, as is, to the cloud; setting up scale parameters and expecting better performance. Auto-scale out/in will not have any implication to a solution that already lags on performance (response time, throughput, etc).

The cloud will be a great host/environment for applications to scale and demonstrate its true potential if, and only if, the solution is optimised to perform well in the cloud.

3. Don't embrace the cloud considering it to be cheap

There is a general myth in the industry that moving to the cloud will bring down costs. There are cost benefits for sure, but economics is a relative term, and it all depends on the solution landscape chosen in the context of the cloud. In fact, benefits like just-in-time provisioning of compute workloads in lieu of procuring them independently, the choice of the compute environment, access to managed platforms, auto scaling or the elastic nature of the cloud

far outweigh just the cost factor.

In summary, embrace the cloud for factors beyond cost, while cost benefits are realised over a period of time.

4. Don't just 'lift and shift'

If you already have your solution hosted on premise or with a third party, taking equivalent boxes and hosting workloads is no different than your current state. We term this as 'lift and shift'. While you can do this to ensure your solution works well in the new environment, a desirable state to be in is to optimise your workloads to leverage the power of the cloud.

Preparation is key to any activities under the umbrella of 'legacy modernisation'. A business or enterprise that wishes to embrace the cloud should not attempt a 'lift and shift' model considering the plethora of benefits one gets while optimising for the cloud. While the cloud addresses scale and elasticity as key tenets, there are multiple ways to solve a given problem/scenario.

Hence, instead of pure play 'lift and shift', consider the following recommendations:

- Understand and evaluate the various options to modernise your solution, while being aware of the available services in the cloud
- Assess the economic implications of choosing one over the other
- Make the required changes to your solution in view of the option chosen
- Migrate your solution

5. Latency exists

Cloud regions are evolving and, hence, the eventual choice a customer makes is to host workloads at the nearest data centre. Having deployed workloads on premise and then moving to the nearest available region (outside of self-hosting) definitely instills certain latency. Geographic proximity, agreement with local telecom/network services providers in optimising routes, etc, will result in less latency. However, when a public cloud is chosen, irrespective of whether it is in the same region or the nearest region, you will notice a certain level of latency.

Do not take it for granted that latency exists, as there are ways and means to address this issue while you move your workloads to the cloud. Certain design changes leading to optimisation may be required to ensure your solution works optimally in the new environment.

6. Transient failures

Unlike on-premise systems that may run physically on a corporate LAN or within a defined boundary of infrastructure, things are rather different in the cloud. Due to the distributed nature of the infrastructure hosting the cloud, and with the need to subsume transitory issues, the infrastructure would, for example, need to swap an instance facing a problem with